# Nicolò Marmi

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## EDUCATION

**Stony Brook University** | Stony Brook, NY **August 2022-December 2026**

*Honors Bachelor of Science, Computer Science Major, Applied Mathematics & Statistics Major*

- Cumulative **GPA: 3.91/4.0** | Dean's List Scholar | Presidential Merit Scholarship
- Computer Science Honors Program

**Relevant Coursework:** Data Structures and Algorithms, Honors Analysis of Algorithms, Machine Learning, Software Engineering, Finite Mathematical Structures, Probability and Statistics, Data Analysis.

## TECHNICAL SKILLS

**Languages:** Python, Java, JavaScript/TypeScript, C, C++, SQL
**Frameworks & Libraries:** PyTorch, Scikit-learn, Pandas, NumPy, OpenCV, React, Next.js, Express.js, TA-Lib, Polars
**Tools & Platforms:** Git, Docker, MongoDB, REST APIs, Linux/Unix, AWS (EC2/S3), Vercel

## WORK EXPERIENCE

**Fineco Asset Management** **Dublin, Ireland**

*Quantitative Developer Intern* May 2026-August 2026

- Developing a TCN-LSTM hybrid model in PyTorch to detect high volatility regime transitions on 26 years of SPX daily and intraday data.
- Engineered the data and feature pipeline, labeling volatility regimes and constructing training sequences; designing the evaluation framework to benchmark model recall and accuracy across market conditions.

**Capco** **Milan, Italy**

*Software Engineer Intern* April 2024-October 2024

- Built the translation layer (OpenAI API, Pydantic validation) for a pipeline enabling UniCredit to collect branch-employee feedback across 2,000+ branches for senior management.
- Replaced an $O(N^2)$ post-processing approach with a two-pass $O(N)$ algorithm, mapping unstructured model output onto a hierarchical tag schema nested several levels deep.
- Reduced API costs 13% by identifying and eliminating redundant system prompts.
- Refactored modules of LORE ML explainability framework, improving code clarity and maintainability. [1]

## PROJECTS & ACTIVITIES

**Quantum ML** **Stony Brook, NY**

*Honors Research Thesis* January 2026-December 2026

- In collaboration with Brookhaven National Laboratory, designed a two-stream neural network in PyTorch to predict unknown reduced density matrices from known marginals of the same quantum system.
- Reconstructed quantum states from partial measurements, reducing tomography cost and enabling more efficient quantum simulations.
- Achieved 8% average MSE on 8-qubit circuit marginal reconstruction, outperforming the benchmark full-tomography approach on the same task with reduced computational cost.

**Algorithmic Trading & Backtesting System** | *Python, Polars, TA-Lib, Git* **Remote**

*Software Engineer, Algorithms Engineer* August 2024-February 2025

- Built the analytics engine for a 0DTE SPX options backtesting system: cashflow and notional-normalized returns, win/loss distributions, Kelly sizing, and VIX-conditioned performance across thousands of trades.
- Extended the strategy engine with intraday stop-loss and take-profit triggers, including same-day collision handling, and eliminated a look-ahead bias that was evaluating exit conditions on bars preceding position entry.
- Adapted Connors' strategies, optimizing parameters to evaluate performance under varied market regimes [2],[3].

**DraftIQ + Player Data API** | *JavaScript, Node.js, Next.js, MongoDB Atlas, Git* **Stony Brook, NY**

*Creator* January 2026-May 2026

- Led a team of 4 building DraftIQ, a full-stack fantasy baseball draft application, and a player data API providing MLB statistics, analytics, and draft recommendations.
- Engineered a scalable backend using Express and MongoDB aggregation pipelines with async request handling, optimized for low-latency queries.